Example 3.11 The parallel circuit shown in the Fig. 3.37 is connected across a single-phase 100 V, 50 Hz ac supply. Calculate:

(i) the branch currents

(ii) the total current

(iii) the supply power factor

(iv) the active and reactive power supplied by the source.

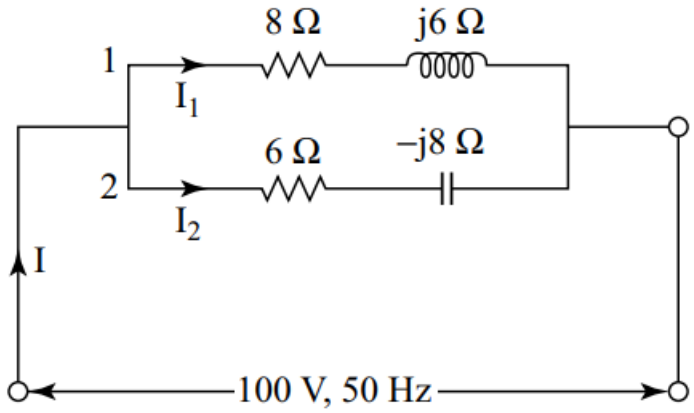


Figure 3.37

It is assumed that the students are aware of the method of representation of a complex number in the forms of a + ib or a + jb. However, this has been explained in the next section.

$$Z_1 = R + jX_L = 8 + j6 = \sqrt{8^2 + 6^2} \angle \tan^{-1}\frac{6}{8} = 10\angle 40°$$

$$Z_2 = R - jX_C = 6 - j8 = \sqrt{6^2 + 8^2} \angle -\tan^{-1}\frac{8}{6} = 10\angle -48°$$

$$I_1 = \frac{V}{Z_1} = \frac{100\angle 0}{10\angle 40°} = 10\angle -40°\text{A}$$

$$I_2 = \frac{V}{Z_2} = \frac{100\angle 0}{10\angle -48°} = 10\angle 48°\text{A}$$

$$I = I_1 + I_2 = 10\angle -40° + 10\angle 48°$$

$$I = 10\cos 40° - j10\sin 40° + 10\cos 48° + j10\sin 48°$$

$$= (10\cos 40° + 10\cos 48°) + j(10\sin 48° - 10\sin 40°)$$

$$= 10 \times 0.766 + 10 \times 0.669 + j(10 \times 0.743 - 10 \times 0.642)$$

$$= 7.66 + 6.69 + j(7.43 - 6.48)$$

$$= 14.35 + j0.95$$

$$= \sqrt{(14.35)^2 + (0.95)^2}$$

$$= 14.45\text{ A}$$

Power factor angle, $\phi = \tan^{-1}\frac{0.95}{14.35} = 4°$

Power factor, $\cos\phi = 0.99$

Active power $= VI\cos\phi$

$= 100 \times 14.45 \times 0.99$

$= 1330\text{ W}$

Reactive power $= VI\sin\phi$

$= 100 \times 14.45 \times 0.069$

$= 99.7\text{ VAR}$

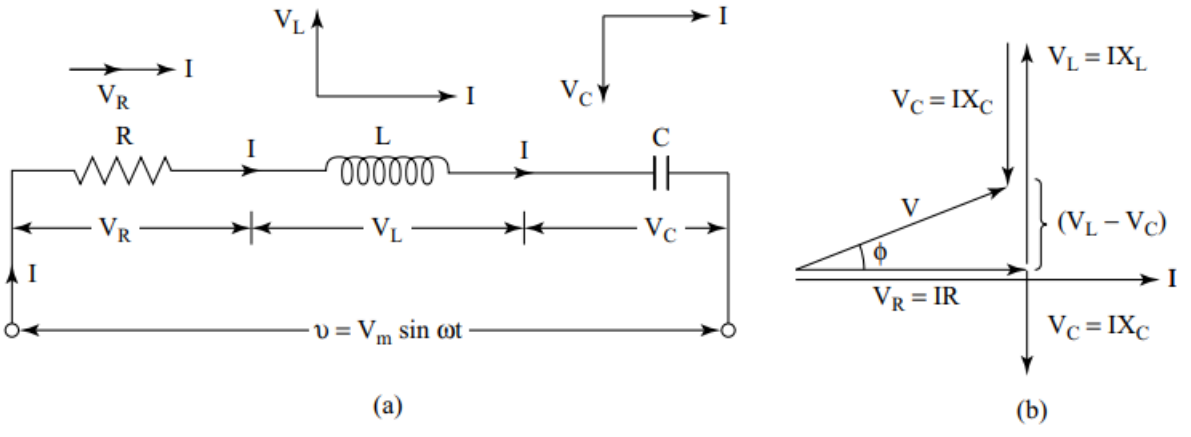


Figure 3.25 *(a) R–L–C series circuit; (b) phasor diagram*

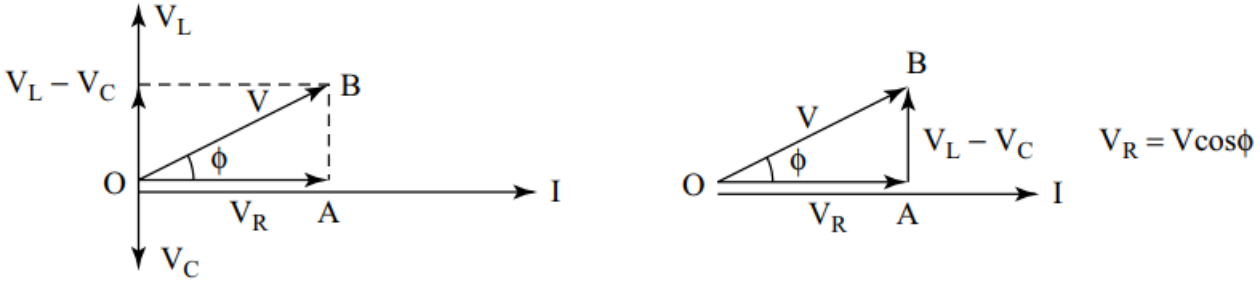


Figure 3.26 *Phasor diagram of current and voltage drops in an R–L–C circuit where $X_L > X_C$*

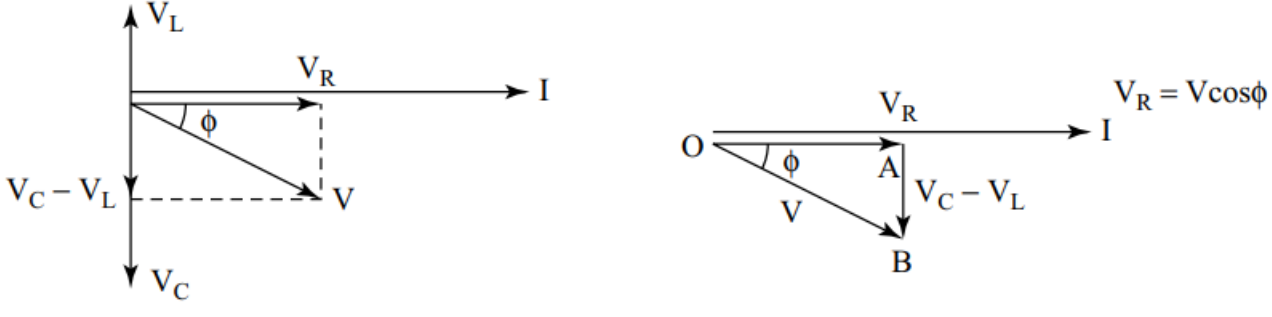


Figure 3.27 *Phasor diagram of an R–L–C series circuit when* $X_L < X_C$